

QF600 Asset Pricing — Week 3 Reference Cheatsheet

Capital Asset Pricing Model: beta, alpha, the market portfolio, SML vs CML, risk decomposition

Built from Lecture 3 slides (Liu Peng, LKCSB) · September 2026

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Source: "Lecture slides - Week 3 v5" (Liu Peng, LKCSB, 55 slides). Slide numbers are PDF page numbers, shown as [S#]. Material not on the slides is tagged **[Beyond slides]**. △ marks a slide

ambiguity or error, with the corrected version. Several sections are marked “(optional)” on the slides; they are still covered here, and noted.

Exam context (Week 1 [S5]): closed book, one double-sided A4 sheet, 40% MCQ and 60% written (concepts plus simple calculations), **no coding**. In the exam you need the CAPM/SML equation, $\beta = \text{Cov}/\text{Var}$ and its derivation, the variance split, portfolio β , CML vs SML, α interpretation and sign logic (negative β), and the equilibrium argument (tangency = market).

Notation used throughout (slide conventions, [S13] footer)

Symbol	Meaning	Dim / units	Time index	Notes
$\tilde{R}_i$	random (realised) return on asset i	per period (% , decimal)	t	Tilde = random
μ_i	expected return $E[\tilde{R}_i]$	per period	—	The quantity being priced
R_f	risk-free rate	per period	known at the start of the period	“Reward for time, or waiting”
$\tilde{R}_M, \mu_M$	market portfolio return, and its mean	per period	t	Slides also write R_m, σ_m
$\mu_M - R_f$	market risk premium (MRP)	per period	—	$\approx 7.5\%/yr$ on the slides [S17, S46]
β_i	beta: sensitivity to the market (Cov/Var, 1.7)	dimensionless	constant	
α_i	alpha: pricing error / abnormal return; CAPM predicts 0	per period	constant	Jensen (1968)
$\tilde{\varepsilon}_i$	idiosyncratic shock	per period	t	$E = 0,$ $\text{Cov}(\tilde{R}_M, \tilde{\varepsilon}_i) = 0$
$\sigma_i^2, \sigma_M^2, \sigma_{\varepsilon i}^2$	total, market, and idiosyncratic variance	return ²	—	The slides write both σ_{ε}^2 and $\sigma_{\varepsilon i}^2$
R^2	systematic share of variance, $\beta_i^2 \sigma_M^2 / \sigma_i^2$	[0,1]	—	
w_i (or w_{pi})	weight of asset i in portfolio p	fraction	—	
β_p, μ_p, σ_p	portfolio beta, mean, volatility	—	—	
μ, Σ, e	N -vector of means, covariance matrix, vector of ones	—	—	Week 2 notation
$w_{tg}, \mu_{tg}, \sigma_{tg}$	tangency portfolio weights, mean, volatility	—	—	

Symbol	Meaning	Dim / units	Time index	Notes
a	fraction of wealth in the tangency portfolio (CML)	scalar	—	$1 - a$ in cash
λ	scalar in the tangency FOC $\Sigma w = \lambda(\mu - R_f e)$	scalar	—	⚠ Not risk aversion and not a factor premium (W4); just a proportionality constant
H_S	$(\mu - R_f e)' \Sigma^{-1} (\mu - R_f e) = \text{squared maximum Sharpe ratio}$	dimensionless (per period)	—	CML slope $= \sqrt{H_S}$

0. Topic map

PREREQUISITES

- W1 Expected utility, risk aversion → mean-variance preferences (quadratic utility or normal returns)
W2 Frontier, tangency portfolio $w_{tg} \propto \Sigma^{-1}(\mu - R_f e)$, CML $\mu_p = R_f + \sigma_p \sqrt{H_S}$, two-fund separation, estimators $\hat{\mu}$, $\hat{\sigma}^2$



WEEK 3: THE CAPM

- Discussion: what is pricing? why do we need it? [S2]
- FOUNDATIONS
 - Pricing equation $\mu_i = R_f + \beta_i(\mu_M - R_f)$ [S3]
 - β intuition: >1 amplifies, $=1$ tracks, <1 lags; " β = the ride" [S4–S5]
 - Regression form $\hat{R}_i - R_f = \alpha_i + \beta_i(\hat{R}_M - R_f) + \varepsilon_i$; α = abnormal return [S6–S8]
 - Kodak vs ASML parable; " β is the ride, α is the excess" (Jensen 1968) [S9–S10]
 - Bridging: take $E[\cdot]$ → impose $\alpha = 0$ → SML [S11]
 - Key results sheet: SML, $\beta = \text{Cov}/\text{Var}$, $\beta_M = 1$, $\beta_p = \Sigma w \beta$, σ^2 split [S13]
- BUILDING THE CAPM in 4 steps [S14–S21]
 - R_f (horizon/currency matching) [S14–S15]
 - MRP; market portfolio (value-weighted, unobservable → proxy) [S16–S17]
 - β (0.5–1.7 across industries); $\beta = \text{Cov}/\text{Var}$ by least squares [S18–S20]
 - Full equation; in practice add α and test $\alpha = 0$ [S21] · Quiz Q1–3 [S22]
- PART 1 One portfolio for everyone [S23–S31]
 - W2 recap: tangency FOC, CML, H_S [S24]
 - + market clearing → tangency = market → SML [S25]
 - Sharpe's move (1964; Lintner 1965; Mossin 1966) [S26]
 - Assumptions: two lists of five; homogeneous expectations = load-bearing wall [S27–S28]
 - Everyone picks the same point → M; chart relabelled [S29–S30] · Quiz Q4–6 [S31]
- PART 2 More on beta [S32–S43]
 - Everything moves together (10 industries, 2004–2013) [S33]
 - Systematic (priced) vs idiosyncratic (not priced); Var split [S34]
 - Naming β ; pricing = one multiplication; measuring cup [S35–S38]
 - β as a visible slope: Manuf $\beta = 1.17$, $R^2 = 0.91$ [S39]
 - Three facts: $\beta_M = 1$, $\beta_p = \Sigma w \beta$, diversification can't go below min β [S40]
 - Warning shot: Manuf $R^2 .91$, HiTec .84, Hlth .60, Utils .40 [S41]
 - Negative $\beta \Rightarrow \mu < R_f$ (insurance) [S42] · Quiz Q7–9 [S43]
- PART 3 The SML (optional) [S44–S48]
 - SML: intercept R_f , slope MRP, passes $(1, \mu_M)$; drawn, not fitted [S45]
 - Ten industries vs SML (1.5%, 7.5%): Energy +6.1%, Durables –5.0% [S46]
 - CML (σ , efficient portfolios only) vs SML (β , every asset); $\beta \neq \text{volatility}$ [S47–S48]
- PART 4 Splitting risk (optional) [S49–S53]
 - $\sigma_i^2 = \beta_i^2 \sigma_M^2 + \sigma_{\varepsilon_i}^2$, R^2 = systematic share [S50–S51]
 - Diversification: idiosyncratic 8.9% → 3.0% with 10 industries; market 14.6% [S52]
 - Market model: α not in CAPM, M proxied, everything estimated [S53]
- Synthesis: W1 utility → W2 frontier → W3 market & β [S54]; homework: real α ? [S55]

▼
NEXT: W4 multi-factor models (CAPM = K = 1); W5 factor-neutral construction; W6 SDF

1. Concept entries

1.1 Pricing and the CAPM pricing equation [S2-S3, S21]

- **Discussion [S2]:** “What is pricing? Why do we need it?” The slides pose these questions without answers.

- [Beyond slides] *Pricing* means mapping an asset’s risk into a **required (expected) return**, equivalently a discount rate or a price.
- We need it for valuation (discounting cash flows, W9), cost of capital, performance evaluation (α) and capital budgeting.

- **Statement:**

$$\mu_i = R_f + \beta_i(\mu_M - R_f)$$

- μ_i : expected return on asset i , “the quantity we want to price”.
 - R_f : the risk-free rate, “the reward for time, or waiting”.
 - β_i : the asset’s exposure to the risk factor.
 - $\mu_M - R_f$: the market risk premium, “the market’s excess return over the risk-free rate”.
 - “Expected return is a straight line: a baseline reward for time plus a premium for bearing risk.” The single risk factor is the market portfolio.
 - **In words [S21]:** required return = a safe base (R_f , what you earn taking no risk) + this asset’s share of the market’s risk (β_i) $\times$ the market’s reward per unit of risk ($\mu_M - R_f$).
 - In practice you estimate this as a regression and add an intercept, α . “The CAPM’s whole claim is that $\alpha = 0$: every asset should sit exactly on the line.”
 - **Worked example [Beyond slides]:**
 - Inputs: $R_f = 3\%$, $MRP = 7.5\%$, $\beta = 1.5$.
 - $\mu_i = 3 + 1.5(7.5) = \mathbf{14.25\%}$.
 - If your forecast for the stock is only 12%, its α relative to the CAPM is -2.25% . It is overpriced, so sell (1.7).
 - **Interpretation / limits:**
 - $\beta = 0$ gives $\mu_i = R_f$.
 - $\beta = 1$ gives $\mu_i = \mu_M$.
 - $\beta < 0$ gives $\mu_i < R_f$ (1.9).
 - The model is linear in β with no other variable priced.
 - **Traps:**
 - Using μ_M instead of $\mu_M - R_f$ as the multiplier.
 - Adding R_f twice when working in excess returns ($\mu_i - R_f = \beta_i(\mu_M - R_f)$).
 - Horizon mismatch (annual MRP with a monthly R_f).
-

1.2 Beta: intuition [S4-S5, S18-S19, S35]

- **Definition:** beta is “a fund’s exposure to common market risk: the return it earns for simply ‘moving with the market’”. It is “your sensitivity to the market = the slope”.
- **Regimes:**

β	Label	Meaning
$\beta > 1$	amplifies / aggressive	more market risk than the market itself
$\beta = 1$	tracks	exactly the market; “the unit”
$0 < \beta < 1$	lags / dampens / defensive	less market risk
$\beta < 0$	hedge	moves opposite the market (e.g. gold, hedges)

- **Illustrations:**
 - [S19] typical betas are 0.5 (defensive), 1.0 (market) and 1.6 (aggressive).
 - [S35] shows $\beta = 0.5$ “moves half as much”, 1.0 “moves exactly with”, 1.8 “nearly twice as much”.
 - Across industries β ranges **0.5-1.7** [S18].
- **Rule of thumb [S19]:** “If the market moves 1%, a β -asset moves about $\beta\%$ with it.” This is *on average*, i.e. an expected response, not an exact one.
- **Three framing points [S18]:**
 - “A sensitivity, not a size”: when the market moves 1%, how far does this asset move?
 - “One is the market itself.”
 - “A slope you can measure”: regress the asset’s excess return on the market’s.
- **Rhetorical takeaways [S5, S35]:**
 - “Beta is the return you get from riding the market’s tailwind. Beta only lets you survive in the market.”
 - “Beta answers one question and refuses all the others: is this asset riskier than the market, or safer?”
- **Traps:**
 - A β of 1.6 does **not** mean $1.6\times$ the *volatility*. Beta measures co-movement, not total risk (1.10).
 - A stock can have $\beta < 1$ and still be very volatile.

1.3 The regression (random-variable) form and alpha [S6-S10]

- **Definition (market model / regression form):**

$$\tilde{R}_i - R_f = \alpha_i + \beta_i(\tilde{R}_M - R_f) + \tilde{\varepsilon}_i$$

- “Replace expectations with realized returns and you get the equation you actually estimate, and alpha appears.”
- α_i : abnormal return the model can’t explain. **The CAPM predicts it is zero.**
- β_i : sensitivity of the asset’s excess return to the market’s.
- $\tilde{R}_M - R_f$: the market’s realised excess return each period.
- $\tilde{\varepsilon}_i$: zero-mean noise, i.e. diversifiable, asset-specific surprises.
- “If the CAPM holds, every alpha is zero: all expected return is payment for beta. Testing the model means testing whether $\alpha = 0$.”
- **Alpha [S8]:**
 - “The risk-adjusted excess return earned after stripping out the market factor.”
 - “The extra return you win above the market, through sharper insight, judgment, and action.”
 - Picture: your line ($\beta + \alpha$) sits parallel above the market baseline (β), and the vertical gap is α .
 - “Only Alpha builds differentiation.”
- **Kodak vs ASML [S9, marked optional]:** “the same disruption exposed two opposite instincts”.
 - **Kodak (chased β , defend the rent):** see the digital camera prototype (1975, the first digital camera) → choose to protect film economics → result: filed for bankruptcy (Chapter 11, 2012). “Invented digital. Defended film.”

- **ASML (built α , fund the future)**: see the next lithography wave → choose to fund EUV R&D → result: the only EUV-system maker. The EUV bet took ~20 years and billions; ~100% EUV market share; High-NA EUV shipping 2024. “Harvested DUV. Funded EUV.”
- “The differentiator wasn’t foresight. It was the willingness to fund cannibalization.”
- **“Beta is the ride. Alpha is the excess.” [S10]:**
 - In a regression, β is “what you were paid for moving with everything else” and α is “what is left after the market has been accounted for”.
 - Borrowed for a career or a company, β “keeps you alive” (table stakes) and α “is what compounds” (what you saw and acted on before the market priced it in).
 - “Alpha is the only part that accumulates into anything.”
 - Source: Jensen (1968, *JF* 23(2)), “where α first became a number fund managers were graded on”.
- **Jensen’s α (ex post) [Beyond slides formula]:** $\hat{\alpha}_i = \overline{R_i} - \overline{R_f} - \hat{\beta}_i \overline{R_M} - \overline{R_f}$.
 - *Example*: a fund averages 12%, with $R_f = 2\%$, $\beta = 1.1$ and the market averaging 10%.
 - $\alpha = 12 - 2 - 1.1(8) = +1.2\%/yr$.
- **Traps:**
 - α is **not** “return above the market” in raw terms. It is return above what the fund’s β justifies. A $\beta = 1.5$ fund beating the market by 2% can have negative α if MRP > 4%.
 - α is an estimate with a standard error. Test $t = \hat{\alpha}/SE(\hat{\alpha})$ (1.12).
- **Connections:** W4 (α relative to FF3/Carhart); W5 (α as the residual-mean forecast).

1.4 Bridging the two forms; the key-results sheet [S11, S13]

- **Derivation (slide):**
 1. **Estimate:** $\tilde{R}_i - R_f = \alpha_i + \beta_i(\tilde{R}_M - R_f) + \tilde{\varepsilon}_i$ (what you run on data).
 2. **Take expectations:** the noise averages out ($E[\varepsilon] = 0$), giving $\mu_i - R_f = \alpha_i + \beta_i(\mu_M - R_f)$ (the average relationship).
 3. **Impose $\alpha = 0$:** $\mu_i - R_f = \beta_i(\mu_M - R_f)$ (the expectation form, i.e. the theory).
 - “Alpha is the wedge between data and theory: it is exactly what lets real returns disagree with the model.”
- **Key results [S13]:** “the same model in two languages”.

Expectation form (theory)	Random-variable form (regression)
SML: $\mu_i - R_f = \beta_i(\mu_M - R_f)$ (what should this asset earn on average?)	Market model: $\tilde{R}_i - R_f = \alpha_i + \beta_i(\tilde{R}_M - R_f) + \tilde{\varepsilon}_i$ (what happened this month?)
$\beta_i = \frac{\text{Cov}(\tilde{R}_i, \tilde{R}_M)}{\text{Var}(\tilde{R}_M)}$	CAPM prediction: $\alpha_i = 0$ and $E[\tilde{\varepsilon}_i] = 0$
Building blocks: $\beta_M = 1$, $\beta_p = \sum w_i \beta_i$	Total risk splits: $\sigma_i^2 = \beta_i^2 \sigma_M^2 + \sigma_\varepsilon^2$

- **Trap:** $E[\tilde{\varepsilon}_i] = 0$ holds by construction in any OLS regression with an intercept. The *economic* content of the CAPM is $\alpha_i = 0$.

1.5 The risk-free rate and the market risk premium [S14-S16]

- R_f [S14]: “every required return begins from what you could earn taking no risk at all”.
 - **Reward for waiting:** lend to a government and get a known amount back, i.e. payment for time, not risk.
 - **No surprises:** the payoff is fixed the day you buy; its variance is essentially zero.

- **Floor under everything:** no investor accepts less than R_f for a risky bet, so every return is measured from here up.
- **Up close [S15]:**
 - R_f is the only point with $\sigma = 0$, so it lies on the vertical axis.
 - A line from it up to the tangency portfolio M traces the **Capital Market Line**, “the best return available at every level of risk”.
 - **Which rate in practice:**
 - * **Match the horizon:** a 3-month T-bill for short-term decisions; a 10-year government bond yield for a 10-year valuation.
 - * **Match the currency:** USD flows at a US Treasury rate, EUR flows at a German Bund, etc.
 - * **Nearly, not perfectly, risk-free:** governments carry some default and inflation risk; use the closest liquid proxy.
- **MRP [S16]:** $(\mu_M - R_f)$.
 - Why hold risk at all: the market swings, and you accept that only if paid.
 - “The market’s extra pay”: on average the market earns more than the safe rate, and that gap is the reward for one unit of market risk.
 - “**Observed, not assumed**” (estimated from history).
- **Traps:**
 - Using a nominal R_f with real cash flows.
 - A mismatched horizon or currency.
 - Treating the MRP as known: historical estimates have large standard errors [Beyond slides]. With $\sigma_M \approx 15\%$, a 50-year sample gives $SE \approx 15/\sqrt{50} \approx 2.1\%$.

1.6 The market portfolio [S17, S26, S29-S30]

- **Definition:** “hold every risky asset, in proportion to its size” (value-weighted).
 - Illustrative composition: US stocks 40%, international stocks 30%, corporate bonds 18%, real estate 8%, commodities 4%.
 - Properties: $\beta = 1$ (the benchmark for beta); **diversified** (only market risk remains); $\approx 7.5\%$ **risk premium per year**.
 - It sits at the tangency of the CML with the frontier (“everyone holds this”).
- **Unobservable [S17, S53]:** in practice it is proxied by a broad value-weighted index (S&P 500, MSCI ACWI, CRSP, MSCI World), “none is all risky assets”.
 - [Beyond slides] **Roll’s critique (1977):** testing the CAPM is really testing whether the proxy is mean-variance efficient. A rejection may just mean a bad proxy.
- **Why the market is special [S26]:**
 - Markowitz gave the frontier, but “which point?” Information is vast, views differ, and risk appetite is individual, so there is no single answer.
 - **Sharpe’s move:** “Stop trying. Build a simplified ideal world, solve it there, and see what survives.” Simplify until solvable, then ask which conclusions are robust (“the most-used method in economics”).
 - Sharpe (1964), with Lintner (1965) and Mossin (1966). Nobel 1990, shared with Markowitz and Miller.
- **Equilibrium conclusion [S29-S30]:**
 - “Everyone faces the same frontier, so everyone picks the same point.”
 - Same inputs plus the same optimisation give the same tangency portfolio. Add market clearing, and that portfolio must be the **value-weighted portfolio of every risky asset in existence: M** .
 - The chart is relabelled: “the line has not moved” (same CML, same tangency, same arithmetic), “but the claim has” (M is no longer an estimate; it is everything, value-weighted).
 - Payoff: “an unobservable optimisation is now an observable benchmark”.
 - “Everything from here on is one identity, rearranged, starting with how each asset covaries with M .”
- **Traps:**
 - Equating the S&P 500 with M .
 - Thinking investors hold *different* risky portfolios under the CAPM. They differ only in the cash/ M split, i.e. their position on the CML (two-fund separation).

1.7 Beta = Cov/Var: the least-squares derivation [S20]

- **Statement:**

$$\beta_i = \frac{\text{Cov}(\tilde{R}_i, \tilde{R}_M)}{\text{Var}(\tilde{R}_M)}.$$

- “ β is the OLS slope of the asset’s excess return regressed on the market’s excess return.”

- **Derivation (slide, optional; exam-reproducible):**

1. Fit a line: $\min_{\alpha, \beta} E[(y - \alpha - \beta x)^2]$, with $x = \tilde{R}_M - R_f$ and $y = \tilde{R}_i - R_f$.
 2. Set both partial derivatives to zero:
 - $\partial/\partial\alpha: E[y] = \alpha + \beta E[x]$.
 - $\partial/\partial\beta: E[xy] = \alpha E[x] + \beta E[x^2]$.
 3. Eliminate α [steps Beyond slides]:
 - $\alpha = E[y] - \beta E[x]$.
 - Substitute: $E[xy] = E[x]E[y] - \beta E[x]^2 + \beta E[x^2]$.
 - So $E[xy] - E[x]E[y] = \beta(E[x^2] - E[x]^2)$, i.e. $\text{Cov}(x, y) = \beta \text{Var}(x)$.
- Because R_f is a constant, $\text{Cov}(x, y) = \text{Cov}(\tilde{R}_i, \tilde{R}_M)$ and $\text{Var}(x) = \text{Var}(\tilde{R}_M)$ [Beyond slides; with a time-varying R_f this is only approximate].

- **Equivalent forms [Beyond slides]:**

- $\beta_i = \rho_{iM} \frac{\sigma_i}{\sigma_M}$.
- $\beta_M = \text{Cov}(\tilde{R}_M, R_M) / \text{Var}(R_M) = 1$ [S40].

- **Worked examples [Beyond slides]:**

- $\text{Cov} = 0.0288$ and $\sigma_M = 16\%$ ($\text{Var} = 0.0256$) give $\beta = 1.125$.
- $\rho = 0.6$, $\sigma_i = 30\%$, $\sigma_M = 15\%$ give $\beta = 0.6 \times 2 = 1.2$.
- Five months of excess returns (%): $x = (2, -2, 4, 0, 1)$, $y = (2.8, -2.1, 5.1, 0.4, 1.3)$.
 - * $\bar{x} = 1$ and $\bar{y} = 1.5$.
 - * Deviations: $x : (1, -3, 3, -1, 0)$ and $y : (1.3, -3.6, 3.6, -1.1, -0.2)$.
 - * $\widehat{\text{Cov}} = \frac{1.3 + 10.8 + 10.8 + 1.1 + 0}{4} = 6.0$ and $\widehat{\text{Var}}(x) = \frac{1 + 9 + 9 + 1 + 0}{4} = 5.0$.
 - * $\beta = 1.2$ and $\hat{\alpha} = 1.5 - 1.2(1) = 0.3\%/\text{month}$.

- **Traps:**

- Dividing by σ_M instead of σ_M^2 .
- Using $\text{Var}(\tilde{R}_i)$ in the denominator (that is the reverse regression).
- Regressing R_M on R_i (swapped axes gives Cov/σ_i^2).

- **Implementation:**

```
import numpy as np
def capm_regress(ri, rm, rf):
    y, x = ri - rf, rm - rf
    beta = np.cov(y, x, ddof=1)[0, 1] / np.var(x, ddof=1)
    alpha = y.mean() - beta * x.mean()
    e = y - alpha - beta * x; n = len(y)
    s2 = e @ e / (n - 2) # residual variance
    se_a = np.sqrt(s2 * (1/n + x.mean()**2 / ((x - x.mean())**2).sum()))
    r2 = 1 - e @ e / ((y - y.mean())**2).sum()
    return alpha, beta, alpha / se_a, r2
x = np.array([2, -2, 4, 0, 1]); y = np.array([2.8, -2.1, 5.1, .4, 1.3])
print(capm_regress(y, x, 0.0)) # alpha 0.3, beta 1.2
```

Pitfall: use the same ddof in the Cov and Var (they cancel). Monthly α must be $\times 12$ to annualise.

1.8 From Week 2's two results to the CAPM (tangency = market) [S23-S25, S30]

• Result 1: tangency portfolio [S24]:

- Maximise the risky-only Sharpe ratio: $\max_w \frac{(\mu - R_f e)' w}{\sqrt{w' \Sigma w}}$.
- FOC ("marginal return tracks marginal variance"): $\Sigma w = \lambda(\mu - R_f e)$.
- Solve, then rescale the weights to sum to one:

$$w_{tg} \propto \Sigma^{-1}(\mu - R_f e), \quad w_{tg} = \frac{\Sigma^{-1}(\mu - R_f e)}{e' \Sigma^{-1}(\mu - R_f e)} \text{ [normalisation Beyond slides].}$$

• Result 2: Capital Market Line [S24]:

- Combine cash with the tangency portfolio: $\mu_p = R_f + a(\mu_{tg} - R_f)$ and $\sigma_p = a\sigma_{tg}$.
- Eliminate $a = \sigma_p / \sigma_{tg}$:

$$\mu_p = R_f + \sigma_p \sqrt{H_S}, \quad H_S = (\mu - R_f e)' \Sigma^{-1} (\mu - R_f e).$$

- The CML slope $\sqrt{H_S}$ is the **highest Sharpe ratio any portfolio can reach**, and H_S is the tangency portfolio's **squared** Sharpe ratio.

• To the CAPM [S25]:

- Last week gave the two results.
- Add **market clearing**: everyone shares μ and Σ , so all hold the same tangency portfolio, which must be the market M . Its CML becomes the equilibrium risk-return line.
- Apply the tangency condition asset by asset: " Σ^{-1} collapses to a single number, β ". This gives $\mu_i - R_f = \beta_i(\mu_M - R_f)$ for every asset i .
- "Same mathematics as last week: the CAPM is those two results plus the assumption that markets clear."

• Derivation (the step the slide summarises) [Beyond slides detail]:

1. At $w_M = w_{tg}$, the FOC is $\Sigma w_M = \lambda(\mu - R_f e)$.
 2. Row i of Σw_M is $\text{Cov}(\tilde{R}_i, \tilde{R}_M)$, so $\text{Cov}(\tilde{R}_i, \tilde{R}_M) = \lambda(\mu_i - R_f)$ for every i .
 3. Pre-multiply the FOC by w'_M (with $w'_M e = 1$): $\sigma_M^2 = \lambda(\mu_M - R_f)$.
 4. Divide step 2 by step 3: $\frac{\text{Cov}(\tilde{R}_i, \tilde{R}_M)}{\sigma_M^2} = \frac{\mu_i - R_f}{\mu_M - R_f}$, so $\mu_i - R_f = \beta_i(\mu_M - R_f)$. ■
- The SML is a restatement of the **first-order condition for the market to be the tangency portfolio**. Equivalently, the CAPM holds $\Leftrightarrow M$ is mean-variance efficient.

• Worked example [Beyond slides] (two risky assets; excess means (8%, 5%); $\sigma = (20\%, 15\%)$; $\rho = 0.3$):

- $\Sigma^{-1}(\mu - R_f e) = (1.648, 1.563)$, so $w_{tg} = (0.513, 0.487)$.
- $\mu_{tg} - R_f = 6.54\%$, $\sigma_{tg} = 14.27\%$, Sharpe ratio = **0.458**, and $H_S = 0.210$ ($\sqrt{H_S} = 0.458$ ✓).
- Betas relative to the tangency portfolio: $(1.223, 0.765)$.
- Check: $1.223 \times 6.54 = 8.0\%$ and $0.765 \times 6.54 = 5.0\%$ ✓. The SML holds exactly relative to the tangency portfolio.

• Chart [S30]: the CML runs from R_f ($\approx 1.5\%$) through the star at M ($\sigma \approx 11.7\%$, $\mu \approx 18\%$ p.a., read from the chart), tangent to the risky-assets-only frontier (minimum-variance $\sigma \approx 9.4\%$).

- ⚠ These chart numbers are illustrative and differ from the industry data used later (market $\approx 9\%$ mean, $\sigma \approx 15\%$, [S47]).

• Traps:

- H_S is the *squared* Sharpe ratio. The CML slope is its **square root**.
- The λ in the FOC is a Lagrange/scale constant, not risk aversion.
- The tangency weights need the $e' \Sigma^{-1}(\mu - R_f e)$ normalisation.

• Implementation:

```
def tangency(mu_ex, S):
    z = np.linalg.solve(S, mu_ex)          # Sigma^-1 (mu - Rf e)
    w = z / z.sum()                       # normalise to sum to 1
    HS = mu_ex @ z                        # squared max Sharpe
```

```
beta = S @ w / (w @ S @ w)          # betas vs tangency
return w, np.sqrt(HS), beta
```

Pitfall: if $e' \Sigma^{-1}(\mu - R_f e) < 0$, the “tangency” portfolio is on the inefficient branch (negative weights sum); check the sign.

1.9 The assumptions (“Sharpe’s world”) [S27-S28]

- **List A [S27]** (“a market with no friction at all”, optional; “five assumptions, none is true, together they make the problem solvable”):
 1. **Every investor is rational**: maximises expected utility over mean and variance (Weeks 1-2).
 2. **Information is complete and clean**: same data, same time, no cost.
 3. **Expectations are homogeneous**: same μ and Σ (“the heroic one”).
 4. **Short and long positions are unlimited**: any amount, no borrowing constraint.
 5. **One riskless rate**: the same for lending and borrowing, available to everyone.
 - “Read them again as a list of everything real markets are not.”
- **List B [S28]** (“what each one is doing”, optional; “four are ordinary, one is doing all the work”):
 1. **Price takers**: no single trade moves the price (competitive markets).
 2. **Markets clear**: prices adjust until demand = supply for every risky asset. This is what makes the CAPM **an equilibrium model rather than a portfolio recipe**.
 3. **Everyone optimises mean and variance**: either **quadratic utility or normal returns** will do.
 4. **Homogeneous expectations**: “the load-bearing wall”. If we disagree about μ , we hold different portfolios and nothing survives.
 5. **A single riskless rate**: unlimited borrowing and lending at R_f . “Later we show what happens when we drop this.”
- **△ The two slides give two different lists of “five assumptions”**. List A includes complete information and unlimited shorting; list B includes price-taking and market clearing. Know both. The union is the full set.
- **What breaks when relaxed [Beyond slides]**:
 - **Different borrowing and lending rates** give no single CML, and the model becomes Black’s zero-beta CAPM (intercept = zero-beta return $E[R_z]$ instead of R_f).
 - **Heterogeneous beliefs** mean investors hold different portfolios, so M is no longer each investor’s tangency.
 - **Short-sale limits** can mean the market isn’t efficient.
- **Trap**: “normal returns **or** quadratic utility”, not both. Either makes mean-variance preferences valid.

1.10 Systematic vs idiosyncratic risk; $\beta \neq$ volatility [S33-S35, S48]

- **Observation [S33]**: ten industries and the market, 2004–2013 (cumulative excess return of \$1). “Ten industries, one crisis, one recovery”: every industry falls in 2008 and rises after.
 - Questions posed: is any independent of the rest? If they share one movement, how many numbers do you need per industry? What’s left once you subtract the shared movement, and is anybody paying you for it?
- **Two risks [S34]**:

Systematic risk: **PRICED**

tied to the whole market

shared by every asset

cannot be removed by holding more names

Idiosyncratic risk: **NOT PRICED**

belongs to this asset alone (a fire, a lawsuit, a failed product)

disappears when combined with others

so nobody will pay you to carry it

Systematic risk: **PRICED**
Idiosyncratic risk: **NOT PRICED**

so somebody must be paid to carry it

- **Single-index model [S34]** (optional):

$$\tilde{R}_i = \alpha_i + \beta_i \tilde{R}_M + \tilde{\varepsilon}_i \Rightarrow \text{Var}(\tilde{R}_i) = \beta_i^2 \text{Var}(\tilde{R}_M) + \text{Var}(\tilde{\varepsilon}_i), \quad \text{Cov}(\tilde{R}_M, \tilde{\varepsilon}_i) = 0.$$

- $\triangle$ The single-index model is written in **raw** returns, while the market model [S6, S53] uses **excess** returns.
 - β is the same (constant R_f), but the intercepts differ: raw $\alpha_i^{raw} = \alpha_i^{Jensen} + R_f(1 - \beta_i)$ [Beyond slides].
 - Only the excess-return intercept is the CAPM α .
 - **$\beta \neq$ volatility [S48]**: “Beta is not volatility. It is a share of the market’s volatility.”
 - An asset can be wildly volatile and carry almost no beta if that volatility is its own.
 - “The market pays for co-movement, and nothing else.”
 - **Trap**: ranking assets by σ to infer required returns. The CAPM ranks by β .
-

1.11 Pricing collapses to one multiplication; mispricing; the measuring cup [S36-S38]

- **Statement**: “required risk premium of asset $i = \beta_i \times$ (market risk premium).”
 - Know what the market portfolio is worth, and every other asset follows from its beta.
 - “The problem of pricing every asset in the world has been reduced to the problem of measuring one covariance.”
 - **Trading logic (slide, $\beta = 1.5$)**:
 - The model demands $1.5 \times$ the market premium, “no more, no less”.
 - **Priced above that $\rightarrow$ sell**: the asset offers too little return for its β . Overvalued.
 - **Priced below that $\rightarrow$ buy**: the asset offers too much return for its β . Undervalued, “and the CAPM says the gap should close”.
 - $\triangle$ “Priced above/below” refers to **price**. High price means low expected return, which means negative α . Don’t read it as “expected return above the SML”, which is the opposite (positive $\alpha \rightarrow$ buy).
 - **Measuring cup [S38]**: “Before it, risk was an adjective. After it, risk is a quantity.”
 - Once you can measure something you can price it, trade it, hedge it and regulate it. That is “why a flawed model has survived sixty years”.
 - $\triangle$ [S36] and [S37] are **identical duplicate slides**.
 - **Worked example [Beyond slides]**:
 - Inputs: MRP 7.5%, R_f 3%, β 1.5, so the required return is 14.25%.
 - Analyst forecast 17%: $\alpha = +2.75\%$, underpriced, **buy**.
 - Forecast 12%: $\alpha = -2.25\%$, overpriced, **sell**.
-

1.12 Beta as a visible slope; R^2 across industries; three facts [S39-S41]

- **Manufacturing [S39]**: monthly excess returns vs the market, 120 points (2004–2013).
 - **$\beta = 1.17$, $R^2 = 0.91$** .
 - “When the market moves 5%, manufacturing has moved 5.8% on average” ($1.17 \times 5 = 5.85$ ✓; the chart annotates 5.84%).
 - “The scatter is the rest”, i.e. everything the market doesn’t explain (small here).
 - “ β is a regression slope, which is why anyone with a spreadsheet can compute one.”
- **Warning shot [S41]**: “the same line fits some industries far better than others”.

Industry	β	R^2
Manuf	1.17	0.91
HiTec	1.13	0.84
Hlth	0.68	0.60
Utils	0.54	0.40

- Manufacturing: “the market explains almost everything; CAPM is doing real work.”
- Utilities: “most of what happens here has nothing to do with the market.” The $1 - R^2$ part “is the part of risk nobody pays you for”.
- **Three facts [S40]:**
 1. **Market’s own β :** $\beta_M = \text{Var}(\tilde{R}_M)/\text{Var}(\tilde{R}_M) = 1$, the unit of measurement.
 2. **Portfolio β :** $\beta_p = \sum_i w_{pi} \beta_i$, the weighted average of the holdings’ betas.
 - “Beta adds up. Variance does not.” You get systematic risk without touching Σ .
 - So the CAPM holds for portfolios: $\mu_p - R_f = \beta_p(\mu_M - R_f)$, “free of charge”.
 3. **Diversification cannot help:** “averaging betas cannot take you below the smallest one.”
 - $\triangle$ This is true for **long-only** weights (a convex combination). With short positions (or cash, which has $\beta = 0$), β_p can be anything, including 0 [Beyond slides].
- **Proof that β adds up [Beyond slides]:** $\text{Cov}(\sum w_i \tilde{R}_i, \tilde{R}_M) = \sum w_i \text{Cov}(\tilde{R}_i, \tilde{R}_M)$, which is linear. Variance is quadratic in w .
- **Worked example [Beyond slides]:**
 - Weights (0.5, 0.3, 0.2) on betas (1.2, 0.8, 1.5): $\beta_p = 0.6 + 0.24 + 0.3 = \mathbf{1.14}$.
 - With $R_f = 3\%$ and MRP 7.5%: $\mu_p = 3 + 1.14(7.5) = \mathbf{11.55\%}$.
- **Trap:** a portfolio’s σ is **not** the weighted average of σ s; its β **is** the weighted average of β s.

1.13 Negative beta: an asset worth less than cash [S42]

- **Statement:** nothing stops β going negative.
 - $\beta < 0 \Leftrightarrow \text{Cov}(\tilde{R}_i, \tilde{R}_M) < 0$: the asset rises when the market falls (“gold in some regimes; index puts always”).
 - Then $\beta_i(\mu_M - R_f) < 0$, so $\mu_i < R_f$. You accept a loss on average (relative to cash).
 - “And that is rational: you are buying insurance. The payoff arrives exactly when a dollar is worth most.”
 - “If you have ever paid for insurance, you have knowingly bought a negative expected return. The CAPM prices that instinct.”
- **Worked example (chart):** $R_f = 1.5\%$, MRP = 7.5%, $\beta = -0.45$ gives $\mu = 1.5 - 0.45(7.5) = \mathbf{-1.875\%}$ (the chart shows $\approx -1.9\%$). The market star is at (1, 9%).
- **Interpretation:** a hedge’s low return is the premium paid for insurance. “Expected return below R_f ” is not a mispricing.
- **Connections:** W6 SDF (payoffs in bad states are worth more); W4 (negative loadings lower required returns).

1.14 The Security Market Line and the industry test [S44-S46]

- **SML [S45]** (Part 3, optional): plot β on the x-axis and the CAPM becomes a line, $\mu_i = R_f + \beta_i(\mu_M - R_f)$.
 - **Intercept** R_f : at $\beta = 0$ there is no systematic risk, so the asset earns R_f “whatever its volatility”.
 - **Slope** $\mu_M - R_f$: the price of one unit of β , the same for every asset.
 - **Passes through** $(1, \mu_M)$: the market is on its own line by construction.
 - “The SML is **not a fitted line**. It is a prediction, drawn from the riskless rate and the market alone, before you look at a single asset.”
- **Test [S46]** (ten industries, average returns vs β):

- The line is drawn with **intercept 1.5% and slope 7.5%**, taken before plotting any industry.
- “Nothing sits on it”. Above (positive α , teal on the chart): Utilities, Energy, NoDur, Shops, Hlth, Telcm, Manuf. Below (negative α , red): Durables, Other, HiTec.
- “Energy is **+6.1% a year** above the line, consumer durables **−5.0%** below. For a model that predicts zero, that is a lot.”
- Chart readings (approximate): Energy $\beta \approx 0.97$ at $\mu \approx 14.8\%$, so SML 8.8% and $\alpha \approx +6.0$ ✓. Durables $\beta \approx 1.65$ at $\mu \approx 8.8\%$, so SML 13.9% and $\alpha \approx -5.1$ ✓.
- Pattern [Beyond slides]: low- β industries sit above and high- β below. The empirical SML is **flatter** than predicted, the classic “low-beta anomaly” (Black-Jensen-Scholes 1972; Frazzini-Pedersen 2014).
- **α on the SML:** $\alpha_i = \bar{\mu}_i - [R_f + \beta_i(\mu_M - R_f)]$, the vertical distance to the line.
- **Traps:**
 - Fitting a regression line through the points and calling it the SML.
 - Treating distance from the SML as due to total risk.

1.15 CML vs SML [S47]

- “Different x-axis, different population of points, different meaning for the distance from the line.”

	CML	SML
x-axis	total risk σ	systematic risk β
Equation	$\mu_p = R_f + \frac{\mu_M - R_f}{\sigma_M} \sigma_p$	$\mu_i = R_f + \beta_i(\mu_M - R_f)$
Who is on it	efficient portfolios only (cash + M)	every asset and portfolio , efficient or not
Distance below the line	inefficiency (unrewarded idiosyncratic risk), not mispricing	alpha (mispricing)
Slope	Sharpe ratio of M	MRP

- The chart’s CML runs from 1.5% through the market ($\sigma \approx 15\%$, $\mu \approx 9\%$), so its slope ≈ 0.5 .
- **Chart vs text:** the slide says “a single stock sits far below [the CML]” and labels “individual industries fall well below”. Yet several industries plot **above** the CML: Energy ($\sigma \approx 21$, $\mu \approx 14.8$ vs CML 12.0), Utilities ($\sigma \approx 11.5$, $\mu \approx 10.8$ vs CML 7.3), NoDur and others.
 - In theory **no asset can lie above the CML** if M is the true tangency portfolio.
 - Points above it mean the index proxy was **not ex-post mean-variance efficient** in this sample (Roll’s critique) [Beyond slides].
 - Only Durbl ($\sigma \approx 29$) and Other clearly fall below.
- **Worked link [Beyond slides]:** for an efficient portfolio, $\rho_{pM} = 1$, so $\beta_p = \sigma_p / \sigma_M$ and the SML equals the CML. Any asset with $\rho < 1$ sits below the CML even with $\alpha = 0$.

1.16 Variance decomposition and R^2 [S50-S51]

- **Statement** (Part 4, optional):

$$\sigma_i^2 = \underbrace{\beta_i^2 \sigma_M^2}_{\text{systematic, priced}} + \underbrace{\sigma_{\varepsilon i}^2}_{\text{idiosyncratic, unpriced}}, \quad R^2 = \frac{\beta_i^2 \sigma_M^2}{\sigma_i^2}.$$

- **Derivation ([S51], exam-reproducible):**

1. Single-index model: $\tilde{R}_i = \alpha_i + \beta_i \tilde{R}_M + \tilde{\varepsilon}_i$.
2. Take the variance; the constant α_i drops out: $\sigma_i^2 = \text{Var}(\beta_i \tilde{R}_M + \tilde{\varepsilon}_i)$.

3. The variance of a sum has three pieces: $= \beta_i^2 \sigma_M^2 + \sigma_{\varepsilon_i}^2 + 2\beta_i \text{Cov}(\tilde{R}_M, \tilde{\varepsilon}_i)$.
 4. The cross term is 0: ε_i is the regression residual, orthogonal to the market.

- Result: $\sigma_i^2 = \beta_i^2 \sigma_M^2 + \sigma_{\varepsilon_i}^2$.

- **Also [Beyond slides]:** $R^2 = \rho_{iM}^2$ (since $\beta_i = \rho \sigma_i / \sigma_M$).
- **Chart [S50]:** annual variances (%) for ten industries, split systematic/idiosyncratic, labelled with R^2 .
 - Durbl $\approx$ 840 total (73%), Enrgy $\approx$ 445 (48%), Other (88%), HiTec (84%), Manuf (91%), Telcm (77%), Shops (77%), Hlth (60%), Utils (40%), NoDur (72%).
 - Consistency check [Beyond slides]: $\sigma_M = 14.6\%$ gives $\sigma_M^2 = 213$. Manuf $1.17^2 \times 213 = 292$ vs chart ≈ 305 ; Utils $0.54^2 \times 213 = 62$ vs chart ≈ 65 . Consistent to chart-reading accuracy.
 - Footer: “Two industries, same total risk, completely different split, and the CAPM only pays for the dark part.”
 - $\triangle$ No pair in the chart has exactly equal totals. The closest illustration is **Hlth (≈ 175 , R^2 60%) vs Utils (≈ 165 , R^2 40%)**.
- **Worked example [Beyond slides]:**
 - Inputs: $\sigma_i = 30\%$, $\beta = 1.2$, $\sigma_M = 15\%$.
 - Systematic $= 1.44 \times 0.0225 = 0.0324$; total $= 0.09$; idiosyncratic $=$ **0.0576**.
 - **$R^2 = 0.36$** (so $\rho = 0.6$).
- **Traps:**
 - Decomposing **volatilities** additively. Only **variances** add: $\sigma_i \neq \beta_i \sigma_M + \sigma_{\varepsilon_i}$.
 - Reading a low R^2 as “the CAPM is wrong”. R^2 measures the risk split; the CAPM’s claim concerns α .

1.17 Diversification removes only idiosyncratic risk [S52]

- **Chart:** equally weighted portfolios of the ten industries, annualised idiosyncratic risk vs number held.

N	1	2	3	4	5	6	7	8	9	10
idio. σ (%)	8.9	6.5	5.4	4.7	4.2	3.9	3.6	3.3	3.1	3.0

- One industry: **8.9%** of its own risk on top of the market’s. All ten: **3.0%**, “two-thirds of it gone, for free”. Check: $1 - 3.0/8.9 = 66\%$ of the *volatility*, i.e. $1 - (3.0/8.9)^2 = 89\%$ of the *variance* [Beyond slides].
- The market’s **14.6% never moves**.
- “You cannot be paid for a risk that anyone can delete for free. That single sentence is why only β is priced.”
- **Theory [Beyond slides]:** with N equal-weighted assets and independent residuals of equal variance, $\sigma_{\varepsilon,p} = \sigma_{\varepsilon} / \sqrt{N}$.
 - $8.9/\sqrt{10} = 2.81\%$ vs the observed 3.0%. Slightly higher, because industry residuals are mildly correlated or unequal in size.
 - More generally, $\text{Var}(\tilde{\varepsilon}_p) = \sum_i w_i^2 \sigma_{\varepsilon_i}^2 \rightarrow 0$.
- $\triangle$ “The market’s 14.6% never moves / this floor never moves” is exact only if $\beta_p = 1$. The portfolio’s systematic volatility is $|\beta_p| \sigma_M$. For an equally weighted industry portfolio $\beta_p \approx$ average $\beta \approx 1$, so $\approx 14.6\%$, but a low- β portfolio has a lower floor. What never diversifies away is the $\beta_p^2 \sigma_M^2$ term.

1.18 Measuring β in practice: the market model and its problems [S53]

- **Estimating equation:** $\tilde{R}_i - R_f = \alpha_i + \beta_i(\tilde{R}_M - R_f) + \tilde{\varepsilon}_i$. “The three things that are wrong with it”:
 1. **α is not in the CAPM:** the theory says the intercept is zero, and estimating it is the test.
 2. **M is unobservable, so we proxy it** (S&P 500, CRSP, MSCI World), and none is all risky assets.
 3. **Everything is an estimate:** β and α both have standard errors. “Noisy measurements, not facts.”
- **Testing α [Beyond slides]:**
 - $t = \hat{\alpha}/SE(\hat{\alpha})$, and reject $\alpha = 0$ if $|t| > 1.96$ (5%).
 - *Example:* $\hat{\alpha} = 0.25\%/month$ with $SE\ 0.10\%$ gives **$t = 2.5$** , significant. Annualised $\alpha \approx 3\%$.
- **Other practical issues [Beyond slides]:**
 - Estimation window (60 monthly obs is common).
 - Beta instability over time.
 - Blume/Vasicek shrinkage toward 1 (e.g. $0.67\hat{\beta} + 0.33$).
 - Thin trading biases β downward.
- **Implementation:** see `capm_regress` in 1.7. For rolling betas:

```
import pandas as pd
def rolling_beta(y_ex: pd.Series, x_ex: pd.Series, window=60):
    return y_ex.rolling(window).cov(x_ex) / x_ex.rolling(window).var()
```

Pitfall: pandas rolling cov and var both default to $ddof = 1$, so they cancel. Align the dates first (inner join).

1.19 Synthesis and homework [S54-S55]

- **“Three weeks, one machine” [S54]:**
 - W1 expected utility (how one investor scores a risky choice).
 - W2 Markowitz frontier (which portfolios are worth considering).
 - W3 Sharpe: the market (a benchmark found by general equilibrium).
 - W3 CAPM: β and a price (a required return for every asset).
 - “Markowitz found you the gold: every portfolio worth owning. Sharpe gave you the assay: a standard for judging what any single piece is worth.”
- **Group homework [S55]** (“does your strategy have real alpha?”, due one day before session 4):
 1. **Pick your universe:** any assets and period (stocks, ETFs, bonds, crypto, FX), e.g. 2010–2024 monthly.
 2. **Choose a strategy:** momentum, value, 60/40, risk parity, min-variance, or an MA crossover.
 3. **Backtest via a rolling window:** compute return, volatility, Sharpe and max drawdown vs a buy-and-hold benchmark.
 4. **Test the alpha:** regress $R_p - R_f = \alpha + \beta(R_m - R_f) + \varepsilon$. Is α positive and significant? “ α = your skill, β = just the market.”
- **Metric definitions [Beyond slides]:**
 - Sharpe = $\overline{R_p - R_f} / \sigma(R_p - R_f)$, annualised $\times \sqrt{12}$ for monthly data.
 - Max drawdown = $\min_t (V_t / \max_{s \leq t} V_s - 1)$.
 - *Example:* a path $100 \rightarrow 120 \rightarrow 90 \rightarrow 110 \rightarrow 80 \rightarrow 130$ has $MDD = 80/120 - 1 = -33.3\%$.

```
def max_drawdown(values):
    v = np.asarray(values, float); return (v / np.maximum.accumulate(v) - 1).min()
def sharpe(ex_ret, periods=12):
    return np.sqrt(periods) * ex_ret.mean() / ex_ret.std(ddof=1)
```

- **In-class quizzes:** Q1–3 [S22], Q4–6 [S31], Q7–9 [S43]. The slides show titles only.

2. Comparison tables

2.1 The two forms of the CAPM [S6, S11, S13]

	Expectation form (SML)	Random-variable form (market model)
Equation	$\mu_i - R_f = \beta_i(\mu_M - R_f)$	$\tilde{R}_i - R_f = \alpha_i + \beta_i(\tilde{R}_M - R_f) + \tilde{\varepsilon}_i$
Question	what <i>should</i> the asset earn on average?	what actually happened each period?
Contains α ?	no ($\alpha = 0$ imposed)	yes, estimated
Use	required return, cost of capital	estimating β , testing $\alpha = 0$, performance
Link	take $E[\cdot]$ and impose $\alpha = 0$	

2.2 CML vs SML [S15, S24, S45, S47]

	CML	SML
x-axis	σ (total risk)	β (systematic risk)
Formula	$\mu_p = R_f + \sigma_p \sqrt{H_S} = R_f + \frac{\mu_M - R_f}{\sigma_M} \sigma_p$	$\mu_i = R_f + \beta_i(\mu_M - R_f)$
Population	efficient portfolios only	all assets and portfolios
Below the line means	inefficiency (idiosyncratic risk carried)	negative α (overpriced)
Above the line	impossible in theory ($\triangle$ it happens with proxies)	positive α (underpriced)
Slope	max Sharpe ratio	MRP (price of one unit of β)
Comes from	W2 portfolio choice	W3 equilibrium (market clearing)

2.3 Systematic vs idiosyncratic risk [S34, S50-S52]

	Systematic	Idiosyncratic
Measure	$\beta_i^2 \sigma_M^2$ (share R^2)	$\sigma_{\varepsilon_i}^2$ (share $1 - R^2$)
Source	market-wide	firm-specific (fire, lawsuit, failed product)
Diversifiable?	no; the floor is $\beta_p \sigma_M$	yes; $\sim \sigma_{\varepsilon} / \sqrt{N}$
Priced?	yes	no ("anyone can delete it for free")
Example	Manuf R^2 0.91	Utils $1 - R^2 = 0.60$

2.4 β regimes [S5, S18-S19, S35, S42]

β	Label	Required return	Example
< 0	hedge / insurance	below R_f	gold (some regimes), index puts
0	no systematic risk	R_f	any purely idiosyncratic asset

β	Label	Required return	Example
0-1	defensive, lags	between R_f and μ_M	Utils 0.54, Hlth 0.68
1	the market, tracks	μ_M	index
> 1	aggressive, amplifies	above μ_M	Manuf 1.17, HiTec 1.13, Durbl ≈ 1.65

2.5 The two assumption lists [S27-S28]

List A: “no friction” [S27]	List B: “what each is doing” [S28]	Role / what breaks
Rational mean-variance investors	Everyone optimises mean-variance (quadratic utility or normal returns)	gives the frontier and tangency
Complete, clean, costless information	—	same inputs for all
Homogeneous expectations	Homogeneous expectations	load-bearing: without it, no single tangency = M
Unlimited long/short	—	lets everyone hold the tangency portfolio
One riskless rate (borrow = lend)	Single riskless rate, unlimited borrow/lend	a single CML; drop it → zero-beta CAPM [Beyond]
—	Price takers	competitive markets
—	Markets clear	makes it an equilibrium model: tangency = M

3. Master formula sheet

Pricing [S3, S11, S13, S21, S36]

- $\mu_i = R_f + \beta_i(\mu_M - R_f)$: CAPM / SML (required return).
- $\mu_i - R_f = \beta_i(\mu_M - R_f)$: expected excess return = $\beta \times \text{MRP}$.
- Required risk premium = $\beta_i \times \text{MRP}$.

Regression [S6, S13, S53, S55]

- $\tilde{R}_i - R_f = \alpha_i + \beta_i(\tilde{R}_M - R_f) + \tilde{\varepsilon}_i$: market model. The CAPM predicts $\alpha_i = 0$, and $E[\tilde{\varepsilon}_i] = 0$.
- $\mu_i - R_f = \alpha_i + \beta_i(\mu_M - R_f)$: expectations of the regression.
- [Beyond] Jensen’s $\hat{\alpha} = \overline{R_i - R_f} - \hat{\beta} \overline{R_M - R_f}$ and $t = \hat{\alpha}/\text{SE}$.

Beta [S13, S19-S20, S40]

- $\beta_i = \text{Cov}(\tilde{R}_i, \tilde{R}_M) / \text{Var}(\tilde{R}_M)$; [Beyond] = $\rho_{iM} \sigma_i / \sigma_M$.
- OLS normal equations: $E[y] = \alpha + \beta E[x]$ and $E[xy] = \alpha E[x] + \beta E[x^2]$, giving $\text{Cov}(x, y) = \beta \text{Var}(x)$.
- $\beta_M = 1$; $\beta_p = \sum_i w_{pi} \beta_i$; $\mu_p - R_f = \beta_p(\mu_M - R_f)$.

Risk split [S13, S34, S50-S51]

- $\sigma_i^2 = \beta_i^2 \sigma_M^2 + \sigma_{\varepsilon i}^2$ (cross term $2\beta_i \text{Cov}(\tilde{R}_M, \tilde{\varepsilon}_i) = 0$).
- $R^2 = \beta_i^2 \sigma_M^2 / \sigma_i^2$ [Beyond: = ρ_{iM}^2].
- [Beyond] Equal-weighted idiosyncratic $\sigma_{\varepsilon, p} \approx \sigma_{\varepsilon} / \sqrt{N}$.

Tangency and CML (W2 recap) [S24-S25]

- $\max_w \frac{(\mu - R_f e)' w}{\sqrt{w' \Sigma w}}$ with FOC $\Sigma w = \lambda(\mu - R_f e)$, giving $w_{tg} \propto \Sigma^{-1}(\mu - R_f e)$.
- $\mu_p = R_f + a(\mu_{tg} - R_f)$, $\sigma_p = a\sigma_{tg}$, $a = \sigma_p / \sigma_{tg}$.
- $\mu_p = R_f + \sigma_p \sqrt{H_S}$, with $H_S = (\mu - R_f e)' \Sigma^{-1}(\mu - R_f e)$ (squared max Sharpe).
- Equilibrium: $w_{tg} = w_M = \text{SML}$ [derivation Beyond: $\text{Cov}(R_i, R_M) = \lambda(\mu_i - R_f)$ and $\sigma_M^2 = \lambda(\mu_M - R_f)$].

Homework metrics [S55] [Beyond]

- Sharpe = $\overline{R - R_f} / \sigma(R - R_f)$, $\times \sqrt{12}$ for monthly data.
- MDD = $\min_t (V_t / \max_{s \leq t} V_s - 1)$.

4. Glossary

- **Alpha (α)**: abnormal / risk-adjusted excess return after stripping out the market; the CAPM predicts 0; “the excess”.
- **Beta (β)**: sensitivity to the market, Cov/Var ; “the ride”.
- **CAPM**: Capital Asset Pricing Model (Sharpe 1964, Lintner 1965, Mossin 1966).
- **Capital Market Line (CML)**: the efficient risk-return line from R_f through M in (σ, μ) space.
- **Defensive / aggressive**: $\beta < 1$ / $\beta > 1$.
- **Efficient frontier**: portfolios with the highest return for each risk level (W2).
- **Equilibrium model**: prices adjust until markets clear (vs a portfolio recipe).
- **Excess return**: return minus R_f .
- **Homogeneous expectations**: all investors share μ and Σ .
- **Idiosyncratic (firm-specific, diversifiable, unsystematic) risk**: σ_ε^2 ; not priced.
- **Jensen’s alpha**: the CAPM regression intercept, used to grade managers (Jensen 1968).
- **Market clearing**: demand = supply for every risky asset.
- **Market model**: the regression $R_i - R_f = \alpha + \beta(R_M - R_f) + \varepsilon$.
- **Market portfolio (M)**: value-weighted portfolio of all risky assets; $\beta = 1$; proxied by S&P 500, MSCI ACWI, CRSP, MSCI World.
- **Market risk premium (MRP)**: $\mu_M - R_f$ ($\approx 7.5\%/yr$ on the slides).
- **Max drawdown**: the largest peak-to-trough fall.
- **Price taker**: a trader whose trades don’t move prices.
- R^2 : systematic share of variance.
- **Risk-free rate (R_f)**: the zero-variance return; match horizon and currency.
- **Roll’s critique** [Beyond]: the true M is unobservable, so CAPM tests are tests of proxy efficiency.
- **Security Market Line (SML)**: μ vs β line; intercept R_f , slope MRP.
- **Sharpe ratio**: excess return per unit of σ .
- **Single-index model**: $R_i = \alpha_i + \beta_i R_M + \varepsilon_i$ (raw returns).
- **Systematic (market, undiversifiable) risk**: $\beta^2 \sigma_M^2$; priced.
- **Tangency portfolio**: the max-Sharpe risky portfolio; equals M in equilibrium.
- **Two-fund separation**: every investor holds cash + the tangency portfolio (W2).
- **Zero-beta CAPM** [Beyond]: Black’s version without a riskless asset / with unequal borrowing and lending rates.

5. Self-test

Questions

Q1 (conceptual, easy). Why does the CAPM pay only for systematic risk? Explain why “ β is not volatility” and give an example of a very volatile asset with a low required return.

Q2 (numerical, easy). $R_f = 3\%$, $\text{MRP} = 7.5\%$, $\beta = 1.5$. (a) Required return. (b) An analyst expects 17%. What is α , and buy or sell? (c) Same for an expectation of 12%.

Q3 (numerical, easy). A portfolio holds 50%, 30% and 20% in stocks with $\beta = 1.2, 0.8$ and 1.5 . Compute β_p and the CAPM expected return (R_f 3%, MRP 7.5%). Why can you average betas but not volatilities?

Q4 (numerical). A stock has $\sigma = 30\%$, correlation 0.6 with the market, and the market has $\sigma_M = 15\%$. Find β , the systematic and idiosyncratic variances, and R^2 .

Q5 (derivation). Starting from $\min_{\alpha, \beta} E[(y - \alpha - \beta x)^2]$, derive $\beta = \text{Cov}(x, y)/\text{Var}(x)$. Why does it not matter that x and y are excess returns?

Q6 (numerical, hand regression). Monthly excess returns (%): market $x = (2, -2, 4, 0, 1)$, stock $y = (2.8, -2.1, 5.1, 0.4, 1.3)$. Estimate β and α .

Q7 (conceptual/numerical). $R_f = 1.5\%$, MRP 7.5%. An asset has $\beta = -0.45$. Compute its CAPM expected return and explain why rational investors would hold it.

Q8 (CML vs SML). $R_f = 1.5\%$; market $\mu = 9\%$, $\sigma = 15\%$. Asset A has $\sigma = 25\%$ and $\beta = 0.8$ and is correctly priced by the CAPM. (a) Its expected return. (b) The CML return at $\sigma = 25\%$. (c) Where does A sit relative to each line, and why is that not a contradiction? (d) Its correlation with the market.

Q9 (numerical, testing α). Over 60 months a fund's average excess return is 0.9%/month, the market's is 0.6%/month, and $\beta = 1.2$. $\text{SE}(\hat{\alpha}) = 0.08\%$ /month. Compute α (monthly and annualised) and its t-stat. Is it significant at 5%?

Q10 (numerical, diversification). Each industry has idiosyncratic σ of 8.9%. (a) What does independence predict for an equal-weighted portfolio of 10? Compare with the slides' 3.0%. (b) With $\beta_p = 1$ and $\sigma_M = 14.6\%$, what is the portfolio's total volatility using the 3.0% figure? (c) Correct the statement "the 14.6% floor never moves" for a portfolio with $\beta_p = 0.7$.

Q11 (derivation, hard). Using the tangency FOC $\Sigma w = \lambda(\mu - R_f e)$ and market clearing, derive the SML. Which assumption is "the load-bearing wall", and why?

Q12 (hard, synthesis). Two risky assets have excess means (8%, 5%), $\sigma = (20\%, 15\%)$ and $\rho = 0.3$. (a) Compute the tangency weights and its Sharpe ratio ($\sqrt{H_S}$). (b) Compute each asset's β relative to the tangency portfolio and verify the SML exactly. (c) In the slides' ten-industry test, Energy sits +6.1% above the SML and several industries plot *above* the CML. What two interpretations are available?

Solutions

S1.

- Idiosyncratic risk disappears in a diversified portfolio, so "you cannot be paid for a risk that anyone can delete for free". Only the non-diversifiable part ($\beta^2 \sigma_M^2$) commands a premium [S52].
- β measures co-movement (a share of the market's volatility), not total σ [S48].
- Example: a biotech with $\sigma = 60\%$ but $\rho = 0.1$ with the market ($\sigma_M = 15\%$) has $\beta = 0.1 \times 60/15 = 0.4$. Its required return is only $R_f + 0.4 \times \text{MRP}$.

S2.

- (a) $3 + 1.5(7.5) = \mathbf{14.25\%}$.
- (b) $\alpha = 17 - 14.25 = \mathbf{+2.75\%}$: underpriced (price too low), so **buy**.
- (c) $\alpha = \mathbf{-2.25\%}$: overpriced, so **sell** [S36].

S3.

- $\beta_p = 0.6 + 0.24 + 0.30 = \mathbf{1.14}$, and $\mu_p = 3 + 1.14(7.5) = \mathbf{11.55\%}$.
- Covariance is linear in the weights, so β is too. Variance involves all the pairwise covariances (quadratic in w), so volatilities don't average [S40].

S4.

- $\beta = 0.6 \times 30/15 = \mathbf{1.2}$.

- Systematic = $1.44 \times 0.0225 = \mathbf{0.0324}$; total 0.09; idiosyncratic = **0.0576**.
- $R^2 = 0.0324/0.09 = \mathbf{0.36}$ ($= \rho^2$).

S5.

- The FOCs are $\partial/\partial\alpha: E[y] = \alpha + \beta E[x]$ and $\partial/\partial\beta: E[xy] = \alpha E[x] + \beta E[x^2]$.
- Substitute $\alpha = E[y] - \beta E[x]$: $E[xy] - E[x]E[y] = \beta(E[x^2] - E[x]^2)$, i.e. $\text{Cov} = \beta \text{Var}$. ■
- R_f is a constant, and covariances and variances are unchanged by subtracting constants, so $\text{Cov}(x, y) = \text{Cov}(R_i, R_M)$.

S6.

- $\bar{x} = 1$ and $\bar{y} = 1.5$.
- Deviations: $x(1, -3, 3, -1, 0)$ and $y(1.3, -3.6, 3.6, -1.1, -0.2)$.
- $\text{Cov} = 24/4 = 6.0$ and $\text{Var} = 20/4 = 5.0$, so $\beta = 1.2$ and $\hat{\alpha} = 1.5 - 1.2 = \mathbf{0.3\%/month}$.

S7.

- $\mu = 1.5 - 0.45(7.5) = \mathbf{-1.875\%}$, below R_f (and below zero).
- It is rational because it pays off when the market falls, “exactly when a dollar is worth most”. The negative expected return is an insurance premium [S42].

S8.

- (a) $1.5 + 0.8(7.5) = \mathbf{7.5\%}$.
- (b) CML slope = $(9 - 1.5)/15 = 0.5$, so $1.5 + 0.5 \times 25 = \mathbf{14.0\%}$.
- (c) A is **on the SML** ($\alpha = 0$) but **6.5% below the CML**. That is inefficiency (it carries idiosyncratic risk that is not rewarded), not mispricing [S47].
- (d) $\rho = \beta \sigma_M / \sigma = 0.8 \times 15/25 = \mathbf{0.48}$.

S9.

- $\alpha = 0.9 - 1.2(0.6) = \mathbf{0.18\%/month} \approx \mathbf{2.16\%/yr}$.
- $t = 0.18/0.08 = \mathbf{2.25} > 1.96$, so it is significant at 5%. (With many strategies tested, W4 suggests demanding $t > 3$.)

S10.

- (a) $8.9/\sqrt{10} = \mathbf{2.81\%}$, vs the observed 3.0%. Residuals are mildly correlated or unequal, so diversification is slightly less effective.
- (b) $\sqrt{(14.6^2 + 3.0^2)} = \sqrt{222.16} = \mathbf{14.9\%}$. Almost all of what remains is market risk.
- (c) The systematic floor is $|\beta_p| \sigma_M = 0.7 \times 14.6 = \mathbf{10.2\%}$. It is not 14.6%; what never diversifies away is the $\beta_p^2 \sigma_M^2$ term.

S11.

- In equilibrium everyone holds the tangency portfolio, so market clearing makes it w_M .
- Row i of Σw_M is $\text{Cov}(R_i, R_M)$, so $\text{Cov}(R_i, R_M) = \lambda(\mu_i - R_f)$.
- Pre-multiply by w'_M : $\sigma_M^2 = \lambda(\mu_M - R_f)$.
- Divide: $\mu_i - R_f = \frac{\text{Cov}(R_i, R_M)}{\sigma_M^2} (\mu_M - R_f)$. ■
- The load-bearing wall is **homogeneous expectations**. With different μ or Σ , investors have different tangency portfolios, none needs to equal M , and the derivation collapses [S28].

S12.

- (a) $\Sigma^{-1}(\mu - R_f e) = (1.648, 1.563)$, so $w = \mathbf{(0.513, 0.487)}$. $\mu_{tg} - R_f = 6.54\%$, $\sigma_{tg} = 14.27\%$, **Sharpe = 0.458** ($= \sqrt{0.2100}$).
- (b) $\beta = \Sigma w / \sigma_{tg}^2 = \mathbf{(1.223, 0.765)}$. Check: $1.223 \times 6.54 = \mathbf{8.0\%}$ and $0.765 \times 6.54 = \mathbf{5.0\%}$ ✓. The SML holds exactly against the true tangency portfolio.
- (c) Either (i) genuine mispricing ($\alpha \neq 0$), or (ii) the market proxy is not mean-variance efficient (Roll's critique), in which case the CAPM “test” is a test of the proxy. The flat empirical SML (low β above, high β below) is a known pattern.

Appendix A. Slide coverage checklist

Slide	Content	Section
S1	Title	header
S2	What is pricing? Why?	1.1
S3	CAPM pricing equation	1.1
S4–S5	What is beta; amplifies/tracks/lags	1.2
S6	Random-variable (regression) form	1.3
S7–S8	What is alpha	1.3
S9	Kodak vs ASML (optional)	1.3
S10	β is the ride, α is the excess; Jensen 1968	1.3
S11	Bridging the two forms	1.4
S12	“Introducing CAPM” divider	\$0
S13	Key results sheet	1.4
S14–S15	R_f ; CML sketch; which rate	1.5
S16	MRP	1.5
S17	Market portfolio composition, $\beta = 1$, $\approx 7.5\%$	1.6
S18–S19	β : sensitivity; 0.5–1.7; typical betas	1.2
S20	$\beta = \text{Cov}/\text{Var}$ derivation	1.7
S21	Full equation	1.1
S22	Quiz Q1–3	1.19
S23	Part 1 divider	\$0
S24	Tangency & CML (W2 recap)	1.8
S25	Two results $\rightarrow$ CAPM	1.8
S26	What is the market portfolio; Sharpe’s move	1.6
S27–S28	Assumption lists	1.9, 2.5
S29–S30	Same point; tangency = M	1.6, 1.8
S31	Quiz Q4–6	1.19
S32	Part 2 divider	\$0
S33	Everything moves together	1.10
S34	Two risks; single-index variance	1.10
S35	Naming β (0.5 / 1.0 / 1.8)	1.2
S36–S37	One multiplication (duplicate slides)	1.11
S38	Measuring cup	1.11
S39	Manuf β 1.17, R^2 0.91	1.12
S40	Three facts about β	1.12
S41	R^2 across four industries	1.12
S42	Negative β	1.13
S43	Quiz Q7–9	1.19
S44–S45	SML (optional)	1.14
S46	Ten industries vs SML	1.14
S47	CML vs SML	1.15
S48	β is not volatility	1.10
S49	Part 4 divider	\$0
S50–S51	Variance split & R^2	1.16
S52	Diversification chart	1.17
S53	Market model problems	1.18
S54	Three weeks, one machine	1.19

Slide	Content	Section
S55	Group homework	1.19

Appendix B. Flag summary (⚠)

1. **[S47]** Several industries plot **above** the CML (Energy, Utilities, NoDur...), contradicting “individual industries fall well below”. In theory nothing lies above the CML, so this shows the market proxy is not ex-post efficient (Roll’s critique).
2. **[S52]** “The market’s 14.6% never moves”: the systematic floor is $|\beta_p|\sigma_M$, which equals 14.6% only when $\beta_p = 1$.
3. **[S40]** “Averaging betas cannot take you below the smallest one” holds only for long-only weights. Shorts or cash can bring β_p to 0.
4. **[S27] vs [S28]** give two different lists of “five assumptions”.
5. **[S34, S51]** The single-index model uses raw returns while the market model uses excess returns. Same β , different intercepts ($\alpha^{raw} = \alpha^J + R_f(1 - \beta)$).
6. **[S36] = [S37]** is a duplicate slide.
7. **[S36]** “Priced above $\rightarrow$ sell” refers to price, not expected return. High price means low return means $\alpha < 0$.
8. **[S50]** “Two industries, same total risk”: no exact pair appears; Hlth vs Utils is the closest.
9. **[S24]** H_S is the *squared* Sharpe ratio; the CML slope is $\sqrt{H_S}$. The λ in the FOC is a scaling constant (not risk aversion, not W4’s premium).
10. **[S30] vs [S47]** use different illustrative market numbers ($\mu \approx 18\%$, $\sigma \approx 11.7\%$ vs $\mu \approx 9\%$, $\sigma \approx 15\%$).
11. **[S13]** The idiosyncratic variance is written σ_ε^2 without the asset subscript; it is asset-specific ($\sigma_{\varepsilon i}^2$).